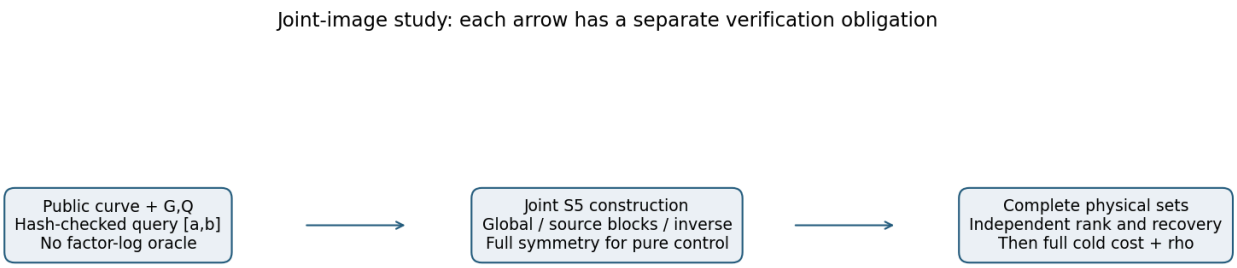
Twist-derived images: joint S5 and attribution controls

This is the continuation of the completed 189-case image-base study and 21 doubling controls (../endomorphism_image_base/README.md). The broad question remains whether twist-derived structure lowers the complete cost of index calculus. Completing a construction or one query does not complete that research goal.



What the completed controls say

The original finite study is complete and is not being repeated. On the same 9,658 conditioned historical residuals, phi produces 1,865 unique signed physical tuples and 1,698 distinct projective augmented rows; doubling produces 1,804 and 1,705. Both attain full rank on all 21 cases. Distinct rows are not independent rank. Through first full rank, the phi arm processes 275,784 finite signed polynomial systems against doubling's 260,496. All 105 complete cold rounds passed their admission checks. Per-case median paired doubling/phi elapsed ratios range from 0.405438 to 5.524357: both gains and losses occur. There is no consistent complete-cost advantage attributable to the phi image in that panel.

The experiments fix one twisted model per case and vary known-log images. They do not compare selecting different twists of a source curve. All targets are public synthetic fixtures with scalar 271; the three seeds are not independent secret-scalar samples. See the original clock boundary (../endomorphism_image_base/BOUNDARY.md) and native summary (../endomorphism_image_base/results/summary.json).

Completed joint construction experiment

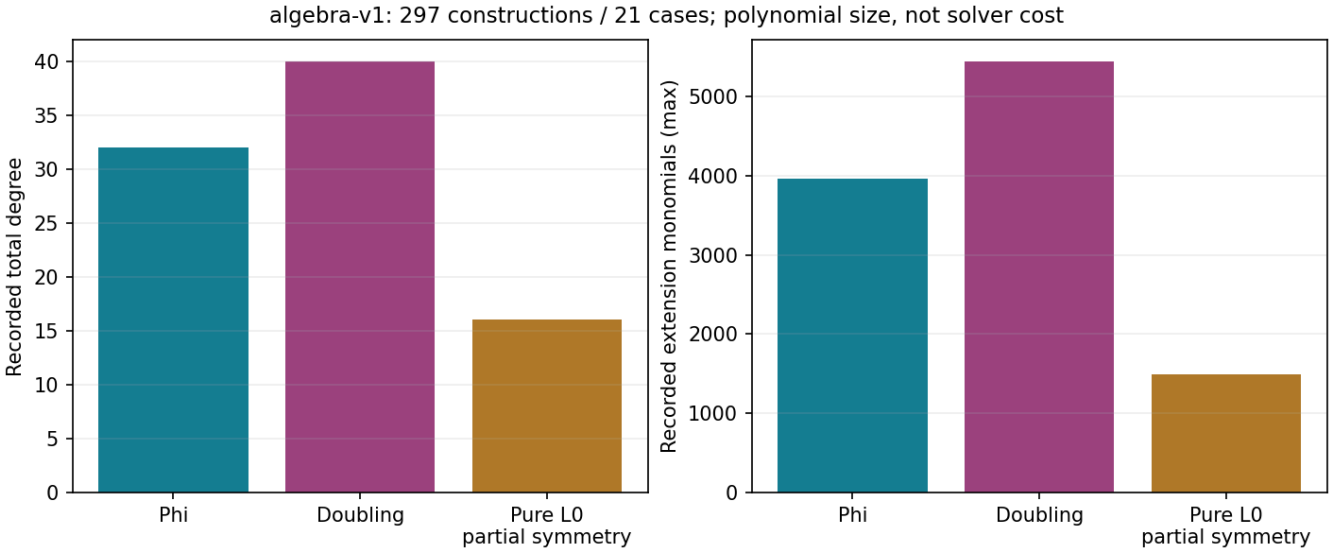
Instead of separately solving each translated S4 system, retain the image source U inside the joint resultant

Res_X(H(e1,e2,e3,X;B), S3(X,Z,x(R);B)).

H is the already verified 49-term S4 expression. Over K=Fp[t]/(t^4-d), B=b0*t^3, the finite-residual construction clears D^8 after substituting Z=N(U)/D(U). Four field components give four equations. Finite-domain constraints enforce actual allowed distinct factors; the geometric encoding uses a discriminant saturation and filters non-base solutions afterwards. R=O has a separately implemented construction, but the 297-input suite does not exercise that branch.

All 297 construction tasks, 694 system files and 63 case-arm sets pass the independent native validator. Each arm has 84 predeclared fixed-query positions, plus separately labelled positive and regression controls. These conditioned controls are not a timing population.

Arm	Tasks	System files	Extension terms	Total degree	U degree
Phi	97	194	3,960	32	24
Doubling	100	200	5,444–5,445	40	32
Pure L0, partial symmetry	100	300	1,484–1,485	16	8



The validator binds the exact workload, public input hashes, coefficient dimensions, term statistics, msolve exports, and zero exit receipts. It checks record consistency and numerical incidence; it is not an independent exact derivation of the generic S5 elimination ideal. Construction source is commit 054b22b41d8e8c0d4296c9119b06e88df5d95fab. A separate whole-stream incidence audit checks 4,110 unique signed physical relations in 5,433 presentations, including actual group sums and numerical S5 vanishing. Every finite construction also has 16 independent Gaussian determinant probes. Incidence of known relations does not establish completeness of a polynomial root finder.

Authoritative numbers and hashes are in algebra-v1-validation.json (results/algebra-v1-validation.json); the construction run report (reports/run-20261024T0003Z/REPORT.md) states its boundary. Lower polynomial degree is a representation result, not a measured solver or IC gain.

Why additional controls are necessary

Three source-block inputs per arm retain all allowed U values in disjoint sets U^12=T. Reduction modulo that equation lowers U degree to at most 11. It does not identify all source phases for a fixed residual or award a twelvefold cost reduction. The source-root partition and map coordinates are checked on all 21 cases; a proof that this partition is the entire folded group action is a separate missing bridge.

The mixed implicit encodings retain a real inverse variable, rather than clearing the denominator. Phi uses U^2*W=1; doubling uses (U^12-B^4)*W=1. Their final polynomials are reduced by the corresponding inverse relation. The pure control instead uses all four elementary symmetric coordinates. Its exact coefficient check removes the auxiliary U and lowers total degree from 16 to 8 on the regression input. Neither generic symmetrization nor an inverse coordinate is a new twist mechanism.

The repaired exporter passes 69 native tests and produces all six finite/geometric regression inputs. The earlier failed export remains preserved. See the repair report (reports/run-20261024T0142Z/REPORT.md). The first six native attempts are terminal: both fully symmetric pure encodings pass their physical-set audits; all four mixed implicit attempts fail allocation under 8 GiB. These are bounded outcomes, not missing true relations or an impossibility proof. Official msolve comparisons must still verify complete signed physical sets and independent rank before a full-cost collector is priced.

All 21 external positions are now reconciled: 11 complete physical-set audits, nine msolve timeouts and one native root-audit timeout after a basis. Every pure source block recovers the same single signed tuple and rank 1/3. Phi's block ranks are 0,2,0; doubling's are 0,0,1. Input reductions and S-pairs vanish for audited bases; reverse ideal containment remains unverified. The terminal report (reports/run-20261024T0841Z-external-terminal/REPORT.md) retains every encoding and failure. These are selected-fixture correctness results, not admitted solver or complete DLP costs. A corrected follow-up reuses terminal failures as well as successes; its earlier duplicate timeout retry and consumed work remain explicitly preserved.

The resumed source-block regression is now complete on the predeclared p37/seed0/query2 fixture. All three unions have complete native roots and pass the separate signed-set/RREF audit: pure L0 has one physical relation and rank 1/3, phi has two and rank 2/3, doubling has one and rank 1/3. The fully symmetric pure finite and geometric variants each recover the same single physical relation and rank 1/3, with independent audits passing. These are one-query correctness and representation results; the phi rank advantage on that selected positive fixture is not a population or cost claim. See the source-block report (reports/run-20261024T0141Z-source-blocks/REPORT.md).

Counting constraint and the zero-residual question

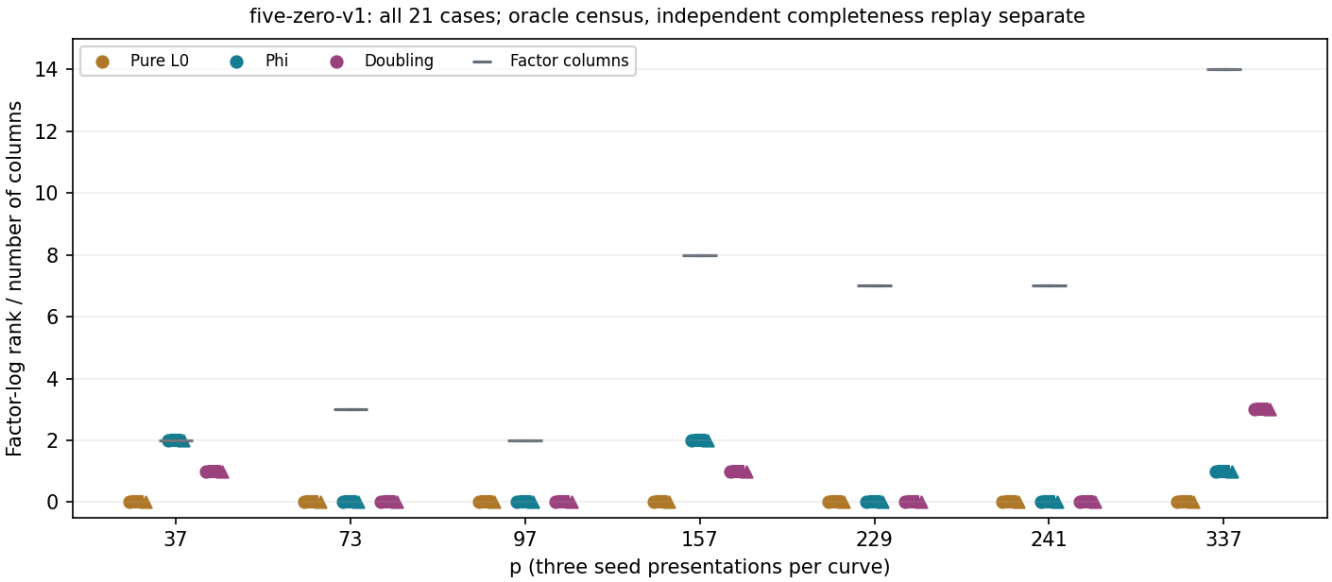
For L ordinary abscissae and a disjoint image of the same size, both mixed arms have exactly 16*binomial(L,3)*L signed physical tuples. Every tuple has exactly one sum in H. Summing relation counts over all residuals is therefore independent of which image map was used. Their uniform-residual means are equal. This does not equate support, variance, projective rows, rank, conditional trajectories or solving cost.

Pure L0 has 16*binomial(L,4) physical tuples. Its partial three-plus-source encoding has four presentations of each tuple. The mixed/pure domain-size ratio 4L/(L-3) is available to generic doubling as well. It cannot be credited specifically to a twist. The old conditioned traces do not estimate an independent uniform-residual mean.

Zero is fixed by the action and might concentrate relations useful for factor-log precomputation. It can also concentrate identities whose entire folded row vanishes. The completed exact diagnostic covers all 21 zero fibers and the entire residual population for p=37, seeds 0,1,2. It records signed physical sets, zero rows, distinct rows and independent RREF. Every zero query has target coefficient zero; no target recovery is claimed from zero-only rows. This is a separate oracle diagnostic, not a polynomial candidate or an IC price. Its protocol is SUPPLEMENT_UNIFORM_AND_ZERO.md (SUPPLEMENT_UNIFORM_AND_ZERO.md).

The full p37 population gives phi and doubling exactly 4,112,640 signed tuples each. Doubling has greater support (1,584,745 residuals versus 1,551,768) and smaller recorded second moment. All 21 four-factor zero fibers have no original, phi or pure relation. Doubling has genuine relations; at p337 its factor-log rank is 1/14, so not every relation is a zero-row identity. See the complete census report (reports/run-20261024T0153Z-uniform-zero/REPORT.md).

The finite-Fp rejection audit preserves every saved physical set, row, rank and recovery. It rejects one phi and two doubling systems over the whole tape, but zero systems before first full rank in either arm. This predicate supplies no through-rank system saving here. The audit report (reports/run-20261024T0607Z-pruning-science/REPORT.md) retains all 21 cases and its exact boundary; no filter is deployed.



The completed five-factor oracle census gives both mixed arms 504 signed physical tuples and 15 distinct projective rows in aggregate. Pure is empty. At p37, phi has factor-log rank 2/2 and doubling 1/2; at p157, 2/8 versus 1/8; at p337, 1/14 versus 3/14. Every seed's result remains separate. Only the p37 phi rows recover all factor logs; no target is recovered from zero-only equations. These are oracle-census observations, with independent point-keyed completeness replay (reports/run-20261024T0908Z-five-point/REPORT.md) now passing all 21 cases and 63 arm comparisons. The field/curve kernel is shared, while point keys, canonical partition and elimination differ. There is no higher-arity polynomial collector or new complete price. See the five-factor report (reports/run-20261024T0802Z-five-zero/REPORT.md).

Verification and completion criteria

Requirement	Status and evidence
Resumed 189 cases and doubling control	Verified complete in the parent study
Joint construction over all fixed inputs	Verified complete: 297 tasks, 694 files
Whole-stream relation incidence	Verified complete: 4,110 physical relations
Native global joint S5 regression	Six terminal results: five allocation failures and one incomplete cooperative result; original limits retained
Source-block regression	Complete on p37/seed0/query2: all three unions and independent signed-set/rank audits pass; ranks 1/3 pure, 2/3 phi, 1/3 doubling
Implicit and fully symmetric regression	Complete: pure finite/geometric audited, each rank 1/3; all four mixed processes fail allocation
Full input population for new variants	Independent native validation passes all 297 positions: 294 new and 3 reused, no missing/invalid/bounded-failure records; run report (reports/run-20261024T0320Z-variant-population/REPORT.md)
External S5 bases and roots	All 21 positions reconciled: 11 physical-set passes, nine external timeouts and one native root timeout; reverse ideal containment remains unverified
Uniform mass and zero-rank diagnostic	Complete native census: all 21 zero cases and three p37 full populations; exact masses, support, histograms, second moments and rank retained
Five-factor zero census	All 21 cases complete; all 63 separate public-point set/rank comparisons pass; equal mixed aggregate counts conceal curve-dependent rank differences
Historical pure S5 miss binding	Corrected checker passes full 13-key/61-query projection, exact tuple and four corruption controls; no retroactive runtime seal or full-stream collector claim
Polynomial identities	Three exact integer/IDC checks and four coordinate Lean theorems pass; factored quartic gate v6 passes seven theorems plus two local AC declarations; all earlier failures retained
Finite-count Lean proof	Repaired module passes the gate and 12-declaration axiom audit
Zero-triangle coefficient identity	Lean theorem and all 21 separate actual-point/set/row checks pass; p337 has 24 additional tuples per seed and rank 1/14

External audit software gate	Version 2 passes 75 tests; block-scope error caught before use; corrected version 3 passes all 76 tests including the actual physical fixture scope test
Pure-zero algebraic certificate	Exact native 2048*D=sum Ai*Ci, two Lean theorems and independent IDC1h09d838b7dca9895f pass; full curve/basis formal bridges remain open
Stronger finite-component obstruction	Exact 64*S^3 certificate, combined 10-theorem Lean gate and independent IDC1h65c11f01d0b1e67a pass; full curve/group bridge remains separate
Finite Fp residual prefilter	All 21 saved-tape audits pass after 77-test gate; sets/rank/recovery preserved, zero rejections through rank; no filter deployed
New cold DLP, GAE and strongest matched rho	Not measured; no new performance ranking
Selecting among different twists	Not tested by these fixed-model image comparisons
Current software gates	Post-merge release library 4,488 pass/119 existing ignores, touched examples 1+3 pass, naming/validation/size gates pass; separate native suite 81/81

The proof map (proof/PROOF_MAP.md) lists hypotheses and missing bridges. The migration reconciliation (MIGRATION_RECONCILIATION_20261024.md) distinguishes terminal work from interrupted work. Managed continuation units and per-phase receipts preserve one thread, 8 GiB and at most 1,200 s; all heavy work retains the repository's busy reservation. Resource failures do not establish a mathematical obstruction.

Before claiming an IC improvement, a viable representation must collect through rank or the frozen cap, verify recovered target and factor logs, and pass a newly frozen complete-cost comparison. Charge lifts, setup, polynomial construction, every failed solve, roots, group checks, matrix work, recovery and verification. Retain A/A, paired rounds, the strongest existing IC baseline, matched rho, external comparisons and 0/0.1/1 GAE. No new leaderboard or benchmark-index row is promoted here.

Stronger ordinary-component rejection rule

The independent first-discriminant replay passes as IDC1h09d838b7dca9895f, including exact integer reconstruction, 32 extra points and two rejected mutations. The stronger search and combined Lean gate additionally prove that the four H4 components force the monic quartic to have at most two distinct roots in odd characteristic. The square-obstruction report (reports/run-20261024T0445Z-square/REPORT.md) gives the exact identity and all assumptions. The combined gate audits 10 theorems with standard foundations only. The stronger independent replay passes as IDC1h65c11f01d0b1e67a, with zero exact integer remainder, 32 extra points and both constant and multiplier mutations rejected.

The proposed consequence rejects finite peeled S4 residuals whose abscissa lies in Fp when the three remaining ordinary factors are distinct. Infinity is retained because CM triangles can sum to O. The frozen audit must preserve every saved physical presentation, unique tuple, projective row, rank, first full-rank query and recovery on the full historical tape for phi, doubling and pure controls. The concrete degree-four B basis is checked separately. No historical collector or relation oracle is repeated. Rejection counts alone will not be called an end-to-end gain or a gain attributable to the twist.

Prior work

Permutation, low-order-torsion and quasihomogeneous invariants are prior methods: Using Symmetries in the Index Calculus for Elliptic Curves Discrete Logarithm (https://eprint.iacr.org/2012/199) and Symmetrized Summation Polynomials (https://doi.org/10.1007/978-3-642-55220-5_3). The invariant coordinates used for descent need their stated field of definition. Joux–Vitse (https://eprint.iacr.org/2010/157) supplies relevant partially symmetrized extension-field methods. Polynomial degree alone does not establish regularity, complexity, novelty or a speed advantage.